

# Statistics and Financial Data Analysis

A17471E1

## Description of the Problem, Data and Tasks

**Setting:** Implied volatility is a measure of the market's expectation of the future volatility of the price of an underlying asset, as inferred from the prices of the corresponding options. It is not directly observed, but rather obtained by inverting the Black–Scholes formula to find the volatility value that, when input into the Black-Scholes model, reproduces the observed market option price. The collection of implied volatilities (IVs) across a range of strikes and maturities forms what is known as the *implied volatility surface*, a fundamental object in derivatives pricing and risk management. For each fixed maturity  $T$ , the collection of implied volatilities for different strikes  $K$  is called the *Implied volatility smile*. For a fixed strike  $K$ , the collection of implied volatilities for different maturities  $T$  is called the *term structure of Implied volatility*. At-the-money (ATM) implied volatility is the implied volatility at that strike which is the same as the asset price<sup>1</sup> ( $K = S_{t_0}$ ).

**Data:** Following datasets are provided:

`callprices.csv`, `impliedvols.csv`, `newquotes.csv`, `tradedvolumes.csv`, `ATMtsIV.csv`.

The dataset `callprices.csv` contains European Call option price quotes of the same asset, for different maturities  $T$  (expressed in units of 1 year) and different strikes  $K$  (expressed in terms of percent, as a fraction  $k := \frac{K}{S_{t_0}}$ , relative to the price of the underlying). Thus, each entry in `callprices.csv` corresponds to a triple  $(k, T, \text{callprice})$ , for prices of call options on the asset for a given strike and maturity. Similarly, each entry in `impliedvols.csv` corresponds to a triple  $(k, T, \sigma_{\text{impl}})$  where  $\sigma_{\text{impl}}$  denotes the implied volatility corresponding to the respective call price. The dataset `tradedvolumes.csv` contains a time series of traded volumes of the underlying and `ATMtsIV.csv` contains time series of at-the-money implied volatilities at different maturities.

**Task:** You have taken over the assignment of a colleague (Jim). Your job is to assess the available solutions, and suggest improvements where appropriate: Call prices for a given asset are quoted in the market at different strikes and maturities (as you can see in the dataset). But *not all* possible maturities are liquidly traded at every given point in time (for some strikes there may not be an observable quote). Jim created a model to interpolate between available option prices, to estimate quotes also for strikes that are not currently quoted in the market, which can help inform a trading desk. Jim has also fitted a model to the time series of ATM implied volatilities for short maturities and a model for traded volumes, both of which need some attention. Your task is to take over Jim's assignment, comment on strengths and weaknesses of his solutions, and create an independent solution of your own that can serve as a benchmark. Also write a short report discussing strengths and limitations of *your* model.

**General Instructions:** The take-home exam consists of a 'Questions Section' which is 40% of the total mark and a 'Modelling Section' which is 60% of the total mark. You should submit only one PDF file with your answers to both the questions and modelling section. Your answers to the 'Questions Section' should include the code used (as text in the pdf submitted). No code is needed in the modelling section, but you need to display graphs (correctly labelled) and explain your conclusions, writing a coherent report on your model assumptions and diagnostics. You should only include graphs which you discuss in the text of your answer. Your answers to the modelling section should be in the form of a report of at most 1500 words and 20 graphs. These limits will be enforced and should be considered *upper limits* (not a guide to expected length).

<sup>1</sup>The price of the underlying on the reference date  $t_0 = 17/7/2020$  is  $S_{t_0} = 100$ .

## Questions Section (40% of the total mark)

### Preparations [6 marks]

- 1.i. Jim has fitted piecewise linear models to volatility smiles from `impliedvols.csv` at the maturities  $T = 0.1$  and  $T = 2$ , and to call prices from `callprices.csv` at the same maturities. Load the data into your notebook and plot the values of call prices for the maturities  $T = 0.1$  and  $T = 2$  against an appropriate range of strikes in the same plot as the model prediction. Also plot the corresponding values of the implied volatility smiles separately.
- 1.ii. Plot the time-series `tradedvolumes.csv` from 17/07/2020 to 17/06/2024 and the series `ATMtsIV.csv` of at-the-money implied volatility for the maturity  $T = 0.1$  for the period '30/11/2015' - '17/07/2020'. Jim has run an Augmented Dickey-Fuller (ADF) test on these time series. Explain the outcome of the test.

### Model Assessment [20 marks]

2. Compute AIC and adjusted  $R^2$  for Jim's interpolation model at  $T = 0.1$   $T = 2$ . Update or refit Jim's solutions if you see room for improvement among piecewise linear models. Explain your answer.

Next it is suggested to seek a *new model*  $f_T^{call}(k)$  for call prices, where  $k = K/S_0$  is moneyness, and  $T$  is an index for the maturity. This is because it became clear that is desirable for our model  $f_T^{call}$  to have finite, positive second derivatives, and the piecewise linear fits in the above solutions can result in curves with discontinuous derivatives.

3. The solution proposed by Jim is to achieve this by fitting a cubic spline approximation to `callprices.csv` at  $T = 0.1$ . Assess the fitted model for optimality using the AIC as an optimality criterion and report your assessment and observations. Make sure to include an *explanation of the AIC criterion*. Compare with the model in Q2. above, including a comparison of respective degrees of freedom.

As more quotes become available (given in `newquotes.csv`) for the observed asset at new strikes for the maturities  $T = 0.1$  and  $T = 2$ , we use these for assessment.

4. Comment on the Cross Validation (CV) routine performed by Jim and perform a 5-fold CV in addition. Comment, which one is preferable in this case and why. Using the 5-fold CV and given all available observations, determine the optimal model complexity<sup>2</sup>. Report the overall MSE of the model with optimal complexity for the smile at  $T = 0.1$ . Justify your answers and compare the outcome with the above models in Q3 and Q2.

### Time Series [14 marks]      We proceed to analyse the market environment and dynamics.

5. Analyse the time-series given in `tradedvolumes.csv` for the period '2014-01-10' - '2017-01-10'. Jim's model of choice for this time-series is an ARMA model, but the fitted model is unavailable. Refit, or suggest a different model, if necessary. Provide appropriate model diagnostics to support your answer.
6. Finally, consider the time series `ATMtsIV.csv` of at-the-money implied volatility for the maturity  $T = 0.1$  and perform diagnostics. Comment whether this time series supports an ARIMA model. Why or why not? Provide clear evidence from the data to back up your conclusion.

## Modelling Section (60% of the total mark)

time series split

(7.) Discuss potential drawbacks of the models considered in 2., 3. and 4. and suggest possible improvements (no need to implement yet) for interpolating between quoted option prices (or equivalently, between quoted implied volatilities). Discuss advantages and shortcomings of considered improvements to decide which ones to implement. (8.i.) Propose and implement a model to be used instead of the available solutions and show, using appropriate criteria, why yours is better. (8.ii.) Discuss any potential drawbacks of your model that one needs to be mindful of, should it be used in production. (9.i) Propose an appropriate time series model which would be suitable for the time series `ATMtsIV.csv` of the  $T = 0.2$  at-the-money implied volatilities. (9.ii) If time permits, propose a way to efficiently simplify the model of the time series that considers the joint movement of *multiple* maturities (textitterm-strucutre) of ATM implied volatilities, given in `ATMtsIV.csv`.

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<sup>2</sup>You may assume that the optimal number of knots is less than  $\sqrt{n_T}$  where  $n_T$  is the number of observable strikes at maturity  $T$ .

## Jim's Notes

### 1.i Implied volatility smile piecewise linear approximation at $T = 0.1$

$$f_{T=0.1}^{iv}(x) = b_0 + b_1x + c_1(x - \kappa_1)_+ + c_2(x - \kappa_2)_+ \quad (1)$$

with  $b_0 = 1.0793$ ,  $b_1 = -0.9534$ , and  $c_1 = 0.5250$ ,  $c_2 = 0.8731$  at knots  $\kappa_1 = 83.333$ ,  $\kappa_2 = 116.667$ .

**Call price** interpolation at  $T = 0.1$  with piecewise linear splines

$$f_{T=0.1}^{call}(x) = b_0 + b_1x + c_1(x - \kappa_1)_+ + c_2(x - \kappa_2)_+ \quad (2)$$

with  $b_0 = 105.343$ ,  $b_1 = -109.173$ ,  $c_1 = 58.6628$ ,  $c_2 = 60.3351$  at knots  $\kappa_1 = 83.333$ ,  $\kappa_2 = 116.667$ .

**Implied volatility** smile piecewise linear approximation at  $T = 2$

$$f_{T=2}^{iv}(x) = b_0 + b_1x + c_1(x - \kappa_1)_+ + c_2(x - \kappa_2)_+ \quad (3)$$

with  $b_0 = 0.4389$ ,  $b_1 = -0.2992$ , and  $c_1 = 0.1174$ ,  $c_2 = 0.2129$  at knots  $\kappa_1 = 83.333$ ,  $\kappa_2 = 116.667$ .

**Call price** interpolation at  $T = 2$  with piecewise linear splines

$$f_{T=2}^{call}(x) = b_0 + b_1x + c_1(x - \kappa_1)_+ + c_2(x - \kappa_2)_+ \quad (4)$$

with  $b_0 = 97.734$ ,  $b_1 = -94.309$ , and  $c_1 = 39.620$ ,  $c_2 = 52.728$  at knots  $\kappa_1 = 83.333$ ,  $\kappa_2 = 116.667$ .

### 1.ii. Augmented Dickey-Fuller (ADF) test, results and conclusions:

ADF results, tradedvolume:  
test statistic = -4.74594  
p-value = 6.89295e-05  
used lags = 6  
nobs = 979  
Decision @ 5%: Reject H0

ADF results, ATMtsIV:  
test statistic = -3.86699  
p-value = 0.00229169  
used lags = 5  
nobs = 1286  
Decision @ 5%: Reject H0

### 3. Fitted cubic spline in truncated power basis for call price interpolation at $T = 0.1$ maturity

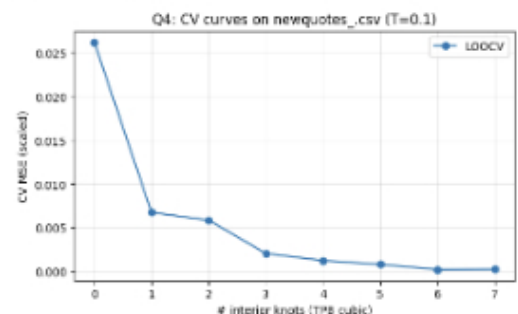
$$f_{T=0.1}^{call}(x) = b_0 + b_1x + b_2x^2 + b_3x^3 + c_1(x - \kappa_1)_+^3, \quad (5)$$

where  $\kappa_1 = 1.039$ ,  $b_0 = 1.0444$ ,  $b_1 = -2.7516$ ,  $b_2 = 2.5508$ ,  $b_3 = -0.7178$ ,  $c_1 = 0.1075$ .

### 4. Leave-one-out cross validation preformed as follows, results shown below:

```
def make_knots_from_quantiles(x_train, k):
    if k <= 0: return []
    qs = np.linspace(0, 1, k+2)[1:-1]
    return list(np.quantile(x_train, qs))

def loo_cv(x, y, k):
    xs = StandardScaler().fit_transform(x.reshape(-1,1)).ravel()
    ys = StandardScaler().fit_transform(y.reshape(-1,1)).ravel()
    knots = make_knots_from_quantiles(xs, k)
    se = 0.0
    for i in range(n):
        n = np.ones(n, dtype=bool); n[i] = False
        res = fit_tpb_cubic(xs[n], ys[n], knots)
        yhat = res['model'].predict(tpb_cubic_design(xs[i:i+1], knots))[0]
        se += (ys[i] - yhat)**2
    return se / n
```



### 5. & 6. Our tests lead to the conclusion that both time-series are weakly stationary. After further diagnostics, we have fitted an ARMA model to the time series of traded volumes, however, the fitted model is unavailable. Diagnostics and fitting of the implied volatility time-series are yet to be performed.